Meat Freshness Detection using E-nose System

Md. Tanvirul Islam
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006095@eee.buet.ac.bd

Md. Mostafizur Rahman
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006096@eee.buet.ac.bd

Rony Ghosh
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006097@eee.buet.ac.bd

Nahian Tasnim
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006098@eee.buet.ac.bd

***Abstract*—This project explores the challenges associated with the perishability of beef, a valuable agricultural product known for its high protein, vitamin, and mineral content, as well as its role in preventing anemia. Despite its nutritional benefits, concerns about health hazards, quality, stability, and safety arise due to its high moisture content. To address these concerns, a freshness monitoring system for food is essential to ensure the consumption of high-quality products. Various techniques, such as spectroscopy, sensory evaluation, computer vision, chemical tests, and electronic nose technology, have been employed for detecting food freshness.**

In this project, we present a cost-effective, portable electronic nose with seven MOS sensors, designed to mimic human olfactory system. This e-nose plays a crucial role in quality control, freshness evaluation, and detection of contaminants in food and beverages. The focus of our research is on developing an e-nose model specifically for assessing the freshness of diverse beef samples. The e-nose comprises a sensing system and an automated pattern recognition system. Analog outputs generated are recorded in an Excel sheet and subjected to analysis using various classification techniques. Notably, the k-nearest neighbors (k-NN) algorithm demonstrates superior accuracy, sensitivity, and specificity. This study lays the groundwork for future enhancements to the model, extending its applicability to assess the quality of various food materials.

***Index Terms*—Beef Freshness, E-nose, Classification, Food Quality Monitoring**

I. Introduction

Consumption of meat, valued for its high protein content and physiological benefits, is on the rise globally, including in Bangladesh. According to the 2017 report from the Organization for Economic Co-operation and Development, the global meat consumption is projected to reach 35.5kg (78.3 lb.) per person by 2024 [1]. As meat consumption continues to grow, the importance of quality in consumer purchasing decisions becomes increasingly evident.

To meet the surging demand, some sellers resort to blending fresh meat with leftover older meat, often stored in refrigerators to prevent spoilage. In an effort to minimize losses, unsold meat is sometimes frozen for extended periods, and small quantities of older meat, such as aged beef, may be mixed with fresh beef without the consumer's knowledge. Beef, being highly perishable due to various post-mortem factors, undergoes rapid quality degradation. The shelf life of uncooked meat in a refrigerator is typically three to five days at temperatures of 14°C or below. Cooking spoiled meat can lead to severe illnesses or food poisoning [2].

Traditional methods for evaluating meat spoilage and quality changes involve microbiological and sensory analyses. While these methods have drawbacks, including destructiveness, cost, and the need for skilled personnel [3], [4], emerging technologies such as electronic nose, computer vision, and artificial tactile sensory technologies aim to replicate human sensory functions in smell, touch, and sight. Electronic noses, though historically considered large and expensive, are now being developed with a focus on being smaller, more affordable, and more sensitive [5], [6].

The electronic nose functions by detecting specific odor components and analyzing their chemical makeup through a pattern recognition mechanism. Although initially applied in food, beverage, and cosmetics industries for quality control, recent applications extend to medical diagnosis for disease-specific odors and environmental protection for detecting pollutants and gas leaks [7].

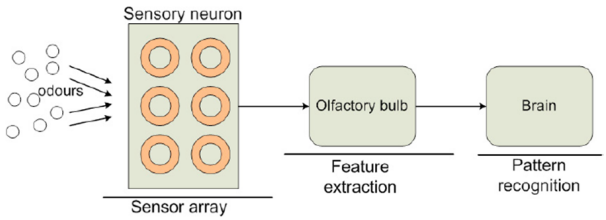


Fig. 1. The functional component of mammalian olfactory system and e-nose.

Food quality assessment can be achieved through supervised or unsupervised learning methods. Supervised learning relies on labeled datasets to train algorithms for accurate classifica-

tion or outcome prediction. Common classification algorithms include linear classifiers, support vector machines, decision trees, and random forests. On the other hand, regression algorithms, such as linear, logistic, and polynomial regression, are used in supervised learning to identify relationships between dependent and independent variables. Unsupervised learning analyzes unlabeled datasets to discover hidden patterns without human intervention.

In this context, we categorize beef into three classes based on time elapsed since collection from the slaughterhouse: Fresh (0-8 hours), Average (8-16 hours), and Rotten meat for the remaining duration. This classification provides a foundation for our investigation into beef freshness using advanced technologies and machine learning approaches.



Fig. 2. Classification Criteria of Meat

II. Literature Review

The initial exploration into utilizing e-noses for assessing beef quality dates back to 2004, as documented by Balasubramanian [8]. A comprehensive array of sensors, comprising 28 gas sensors from a commercial e-nose device, was employed to differentiate between fresh and spoiled beef. The study demonstrated the efficacy of the Quadratic Discriminant Analysis (QDA) classifier in achieving an impressive 98.48% accuracy in solving this binary classification problem.

Subsequent investigations delved into custom e-nose configurations, featuring a reduced set of seven gas sensors accompanied by a temperature and humidity sensor, as conducted by Panigrahi in 2006 [9]. Notably, the QDA classifier maintained its superiority, yielding a classification accuracy of 96%. In the realm of red meat spoilage detection, El Barbri showcased the capabilities of Support Vector Machines (SVM), achieving classification accuracies of 98.81% for beef and 96.43% for sheep meat [10].

The Radial Basis Function Neural Network (RBFNN) found its application in beef freshness detection, attaining an accuracy of 92.2%, as reported by Balasubramanian [11]. A wireless e-nose, employed by Najam ul Hasan, utilized k-NN classifiers to recognize the freshness of both beef and fish [12]. The transition to multiclass classification occurred in 2013, as demonstrated by Papadopoulou [13]. In this study, three distinct sensory classes ('fresh,' 'semi-fresh,' and 'spoiled') were identified. This multiclass approach is deemed crucial for discerning the semi-fresh stage before complete spoilage. SVM emerged as a notable classifier, achieving over 85% accuracy in differentiating among the three classes of beef, as reported by Mohareb [14].

III. Materials and Methods

A. Block Diagram



Fig. 3. Block Diagram of Methodology

In Fig. 3, the block diagram illustrates the configuration of an seven-sensor array forming the E-Nose system. The system includes a sample chamber for housing the acquired beef samples, along with an Arduino Mega. Arduino Mega collects data, with MOS responses acquired as analog output, subsequently stored in an Excel sheet. Subsequently, the accuracy of the collected data undergoes analysis by employing an algorithm through Matlab. In this process, supervised learning is applied to train the model, producing output in the form of confusion matrix, predicting the accuracy of the system's output.

B. E-Nose System

Alias	Sensor	Selectivity
S1	MQ135	Carbon dioxide, alcohol, ammonia, NO_x, smoke, benzene
S2	MQ136	Hydrogen sulfide
S3	MQ2	Alcohol, hydrogen, smoke, Liquefied petroleum gas (LPG), methane, i-butane, propane
S4	MQ4	Methane
S5	MQ6	Iso-butane, propane, LPG
S6	MQ9	Carbon monoxide, methane, and propane
S7	DHT22	Temperature and humidity

Fig. 4. Specifications of the sensors used

C. Working Principle of Gas Sensors

All of these sensors adopt a circular arrangement and operate on the principle of MOS (Metal Oxide Semiconductor) cross-detection. They are designed to detect a mixture of gases effectively. The sensors employ a chemi-resistor, likely composed of tin oxide (SnO2), to detect gases. Tin dioxide, functioning as an n-type semiconductor, governs the conduction current (donor). In a clean air environment, donor atoms (electrons) within tin dioxide are drawn towards oxygen, leading to the adsorption of oxygen on the surface of the sensing material and preventing the flow of electric current.

Upon exposure to a combustible or reducing gas, such as carbon monoxide, an oxidation reaction occurs on the surface of tin dioxide (SnO2), resulting in a reduction of

the density of absorbed oxygen on the sensing material's surface. This reduction, in turn, decreases the potential barrier. Consequently, electrons can traverse the diminished potential barrier more easily, causing a decrease in sensor resistance.

D. Circuit Diagram

The schematic diagram was designed in Altium. Here, we connected the outputs of MQ series gas sensors with the analog input pin of Arduino Mega micro-controller unit(MCU), and the output of DH-T22 temperature sensor with the digital input pin of Arduino mega. During data collection, we gave power inputs to the sensors and the MCU.

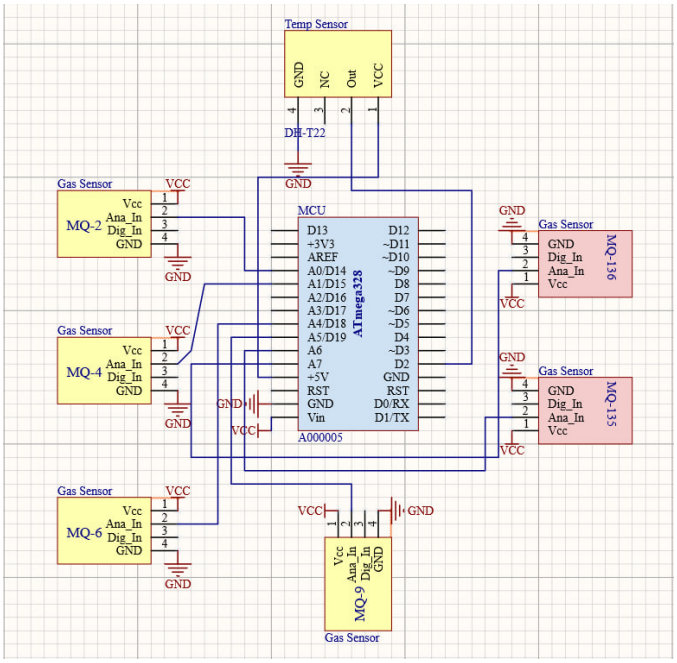


Fig. 5. Circuit Diagram of E-Nose

E. Sample Collection

For sample collection, we collected a fresh meat piece from market. We considered the first 6 hours after slaughtering of the cow as the fresh condition of beef, the next 6 hours as average condition, and the next 12 hours as rotten condition. We shaped the hardware prototype such that it covers the sample container perfectly. Now we took the meat-piece in that container, then covered it with the sensors arrangement. After uploading Arduino code to the MCU and supplying power to the sensors, we stared taking data in an excel file.

F. Experimental Setup

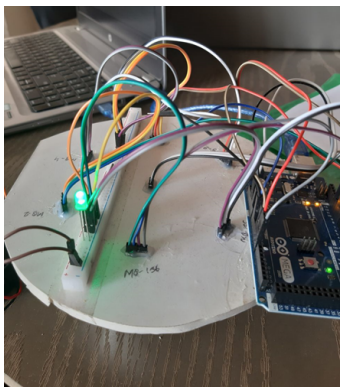


Fig. 6. Hardware Prototype

This image depicts the chamber where the samples were positioned on a holder, directly facing the sensor array. The connections from the sensors were established in accordance with the circuit diagram provided below. Additionally, the load resistors were used which was subsequently linked to the ground. Each sensor's analog output is meticulously connected to the analog input pins of the Arduino Mega, spanning from pin A0 to A7. To ensure a stable power source, a 5V power adapter is employed, supplying both ground and voltage to the system. Thus we created a dataset.

IV. Data Analysis

A. Data Pre-processing

An intricate approach has been taken to pre-process the data, starting with the importation of three distinct datasets: Fresh samples, Average samples, and Rotten samples. The initial step in this pre-processing journey involves the careful removal of noise. Rows in the dataset that fall below or exceed predefined thresholds are examined, and noise is effectively filtered out. Following this, the process delves into indexing and filtering, refining the dataset by retaining rows that meet specific conditions while simultaneously conducting a detailed z-score analysis to eliminate outliers. To enhance data smoothness and reduce variations within the dataset, a moving average filter is applied, emphasizing the improvement of data quality.

Achieving equilibrium among different classes in the dataset is crucial for effective model training. The method adeptly balances the dataset by randomly sampling from each class, ensuring a fair and unbiased representation. Randomizing the order of rows is imperative to prevent biases stemming from the initial data arrangement, and thus, the method carefully shuffles the dataset, fostering a more robust and unbiased analysis.

The dataset is thoughtfully partitioned into training and testing sets through a holdout method, strategically facilitating subsequent training and evaluation phases. Ultimately, the processed training and testing datasets are exported to distinct Excel files. This strategic action guarantees that the processed

datasets remain readily available for further analysis within the ongoing project.

B. Feature Extraction

In the context of our analysis, we have derived 13 distinct features from each sensor within the dataset. These features encompass statistical measures such as Mean, Mode, Median, Standard Deviation, Skewness, Kurtosis, Absolute Difference, Percentiles (25, 50, 75), RMS, and Geometric Mean.

These features collectively provide us with valuable insights into the dataset. They offer measures of central tendency, identify frequently occurring values, mitigate the impact of extreme values, reveal the symmetry and dispersion of the data, and capture the compounded growth or rate of change within the dataset. Such a comprehensive set of features plays a crucial role in classification tasks, enabling a clear and informative comparison between the dataset and other data points. This approach enhances our ability to discern patterns and make informed decisions during the classification process.

C. Feature Selection

Feature selection techniques, as a method for reducing dimensionality, find extensive applications in the realm of data mining and knowledge discovery. The primary objective of feature selection is to carefully choose a subset of variables based on their relevance, and it is increasingly recognized as a crucial component in machine learning [15]. In the context of filter methods for feature ranking, various measures, including Laplacian score, Fisher score, mutual information, entropy, among others, can be employed.

In this analysis, we've employed two distinct feature selection methods: the Filter method and the Wrapper Method. These methodologies have proven instrumental in identifying the most optimal set of features extracted from the dataset, ultimately contributing to achieving the highest accuracy in our results. Ultimately, Filter method has been used for final result.

1) Filter Method: In filter feature selection methods, a statistical measure is used to assign a score to each feature. The features are ranked based on the score and the rank denotes the usefulness of each feature for better classification. Once this ranking has been computed, the features are either selected or removed from the dataset and finally a feature subset composed of the best N features is created.

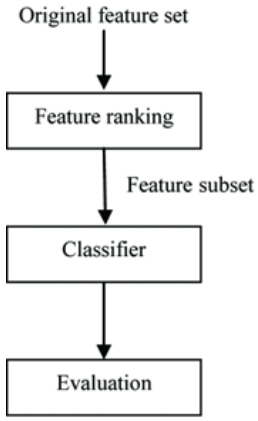


Fig. 7. Flowchart for Filter Meth

2) Wrapper Method: Wrapper methods engage in an exhaustive exploration of the dataset to identify an optimal feature set. A predictive model assesses various feature subsets, assigning scores based on classifier accuracy. While wrapper methods can be computationally demanding, especially for high-dimensional datasets, their superior performance in feature selection is attributed to their consideration of the model hypothesis.

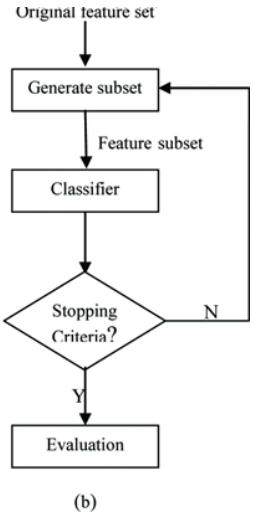


Fig. 8. Flowchart for Wrapper Method

D. Classification

We used two types of classifiers mainly:- k-NN (k- nearest neighbors) and SVM (Support Vector Machine).

1) k-NN: The k-NN model was trained on the dataset of the baseline paper we followed, associating E-nose patterns with

predefined freshness levels of meat samples. For the set of E-nose readings we collected from our meat sample, the k-NN algorithm identified the k nearest neighbors in the training dataset. Freshness level of the test sample was determined by majority voting among its k nearest neighbors. Using this classifier, we found the accuracy to be 82.71%.

2) SVM: SVM classifier determines the optimal hyperplane that maximally separates different freshness levels, creating a decision boundary in the feature space. For the collected set of E-nose readings from a meat sample, SVM predicts the freshness level based on its position relative to the decision boundary. For our experiment, accuracy was obtained 79.21%.

V. Experimental Analysis and Results

A. Confusion Matrix

Confusion matrices depict the aggregate of predicted and actual counts. The term "TN" denotes True Negative, indicating the count of correctly classified negative examples. Similarly, "TP" signifies True Positive, representing the number of accurately classified positive samples. The abbreviation "FP" corresponds to the count of actual negative examples misclassified as positive, while "FN" pertains to the instances of actual positive examples misclassified as negative. Accuracy stands as a widely accepted metric in classification tasks. The formula employed to calculate a model's accuracy, derived from a confusion matrix, is as follows.

Accuracy

The accuracy of the model is calculated using the formula:

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{1}$$

Sensitivity

The sensitivity (True Positive Rate or Recall) is computed as:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

Specificity

The specificity (True Negative Rate) is given by:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{3}$$

The confusion matrix of the project is given below:

TABLE I
Confusion Matrix of the Project

	p-fresh	p-average	p-rotten
t-fresh	36	107	0
t-average	0	228	14
t-rotten	0	0	315

B. Performance Evaluation

Utilizing the confusion matrix facilitates the computation of accuracy, sensitivity, and specificity across various algorithms, constituting the performance matrix. The performance matrix for k-nn algorithms is derived from its confusion matrix.

TABLE II
Performance Evaluation of the Project

	Recall (%)	Precision (%)	F1-Measure (%)	Accuracy (%)
t-fresh	25.17	100	40.22	
t-average	94.21	68.06	79.03	82.71
t-rotten	100.00	95.74	97.83	
Average	73.13	87.93	72.36	

VI. Conclusion

Electronic noses (e-nose) are not universally applicable sensor systems, as they are not equipped to address all gas-sensing applications and may fall short in meeting diverse requirements. This limitation primarily stems from the accuracy and sensitivity constraints inherent in e-nose sensor configurations tailored to specific gases under investigation. Experimental results affirm the safety of consuming beef after 5 days of refrigeration. However, the proposed methodology necessitates a substantial training dataset for optimal results, and the data collection process is time-intensive.

In all scenarios, the efficacy of the proposed algorithm hinges on its robustness in accurately detecting the freshness of beef samples and identifying meat that is unsuitable for consumption. This challenge could potentially be mitigated by augmenting the number of features. The current model considers a single feature for evaluating the dataset, namely the maximum voltage level. To enhance the model, additional features such as the value corresponding to the time taken to reach the peak voltage, the average voltage value among obtained voltages, initial time, etc., can be incorporated.

For future advancements, integrating the electronic nose with Computer Vision (CV) and Artificial Tactile (AT) sensory applications could enhance the practical and reliable detection of meat freshness. Another avenue for improvement involves incorporating Nano gas sensors, where the number of sensors approximates the number of scent receptors in the human nose. This augmentation can significantly enhance the e-nose's ability to discriminate between different aromas.

The proposed model currently attains an accuracy of 82.71%, effectively classifying samples as fresh, average or rotten with minimal errors and within a reasonable computational time-frame. Efforts should be directed towards improving this accuracy to 100% for enhanced overall performance.

Acknowledgment

This project is supported by Department of Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology. Special thanks Dr. Md. Kamrul Hasan (Professor, Dept. of EEE, BUET) and Nawshad Binta Nizam (Adjunct Lecturer, Dept. of EEE, BUET) for their invaluable guidance and support, shaping the success of the project.

REFERENCES

[1] S. Grassi, S. Benedetti, L. Magnani, A. Pianezzola and S. Buratti, "Seafood freshness: e-nose data for classification purposes", *Food Control*, vol. 138, pp. 108994, June 2022.

[2] V. A. Binson and M. Subramoniam, "Exhaled Breath Volatile Organic Compound Analysis for the Detection of Lung Cancer-A Systematic Review", *J. Biomimetics Biomater. Biomed. Eng*, vol. 56, pp. 17-35, March 2022.

[3] S.R. Karunathilaka, Z. Ellsworth and B.J. Yakes, "Detection of decomposition in croaker red snapper and weakfish using an electronicnose sensor and chemometric modeling", *J. Food Sci*, vol. 86, no. 9, pp. 4148-4158, June 2021.

[4] D.R. Wijaya, R. Sarno and E. Zulaika, "DWTLSTM for electronic nose signal processing in beef quality monitoring", *Sens .Act. B: Chem*, vol. 326, pp. 128931, Aug 2021.

[5] V.A. Binson, M. Subramoniam and L. Mathew, "Discrimination of COPD and lung cancer from controls through breath analysis using a self-developed e-nose", *J Breath Res*, Jul 2021.

[6] Binson V A M. Subramoniam, "Design and development of an e-nose system for the diagnosis of pulmonary diseases", *Acta BioengBiomech*, vol. 23, no. 1, pp. 35-44, Jan 2021.

[7] X. Weng, X. Luan, C. Kong, Z. Chang, Y. Li, S. Zhang, et al., "A comprehensive method for assessing meat freshness using fusing electronic nose computer vision and artificial tactile technologies", *J. Sens*, vol. 3, no. 5, Feb 2020.

[8] Balasubramanian, S., Logue, C.M., Marchello, M., 2004. Spoilage identification of beef using an electronic nose system. Transactions of the ASAE 47, 1625–1633.

[9] Panigrahi, S., Balasubramanian, S., Gu, H., Logue, C.M., Marchello, M., 2006. Design and development of a metal oxide based electronic nose for spoilage classification of beef. Sens. Actuat. B 119, 2–14.

[10] El Barbri, N., Llobet, E., El Bari, N., Correig, X., Bouchikhi, B., 2008. Electronic nose based on metal oxide semiconductor sensors as an alternative technique for the spoilage classification of red meat. Sensors 8, 142–156.

[11] Balasubramanian, S., Panigrahi, S., Logue, C.M., Gu, H., Marchello, M., 2009. Neural networks-integrated metal oxide-based artificial olfactory system for meat spoilage identification. J. Food Eng. 91, 91–98.

[12] Najam ul Hasan, Ejaz N., Ejaz, W., Kim, H.S., 2012. Meat and fish freshness inspection system based on odor sensing. Sensors (Switzerland) 12, 15542–15557.

[13] Papadopoulou, O.S., Panagou, E.Z., Mohareb, F.R., Nychas, G.J.E., 2013. Sensory and microbiological quality assessment of beef fillets using a portable electronic nose in tandem with support vector machine analysis. Food Res. Int. 50, 241–249.

[14] Mohareb, F., Papadopoulou, O., Panagou, E., Nychas, G.-J., Bessant, C., 2016. Ensemblebased support vector machine classifiers as an efficient tool for quality assessment of beef fillets from electronic nose data. Anal. Methods 8, 3711–3721.

[15] L. Yu and H. Liu, "Efficient feature selection via analysis of relevance and redundancy", *Journal of Machine Learning Research*, vol. 5, pp. 1205-1224, 2004.